

Assignment 1: Scripps Pier Temperature and Marine Heatwaves, Due 10 October 2024



Figure 2. (a) The original 1,000-ft (305 m) Scripps Pier (right) built in 1916, alongside the new 1,084-ft (330 m) pier during construction, 1988. (b) Pier sampling well, Claude W. Palmer, 1949. (Photographs from SIO Photographic Laboratory, UC San Diego Digital Collections).

Figure from Rasmussen et al. (2019)

Daily measurements of near surface temperature have been collected from the SIO Pier since 1916, resulting in one of the longest continuous ocean temperature time series in the Pacific. The pier time series has been used to illustrate how marine heatwaves have become more prevalent due to anthropogenic warming (Oliver et al. 2018, Fugo et al. 2019). We will perform a similar analysis, focused on the hottest summer months, to examine how extreme ocean temperatures off the pier have changed over the century.

Assignment:

Load the mat file `SIO_TEMP_1917_2023_fill.mat`. The time series from 1917 through 2023 are from the site <https://shorestations.ucsd.edu/data-sio/>. The file variables are:

T: daily near surface temperature ($^{\circ}\text{C}$) at $\sim 0.5\text{-m}$ depth
time: time in Matlab datetime format

- 1) Plot T and indicate extreme high values on the plot, i.e., values that exceed the 99th percentile of the entire record, and are less than the 1st percentile (**Figure 1**). Create histograms showing the counts of extreme values (high and low) as a function of year (**Figure 2**) and month of the year (**Figure 3**).

Q1: When do extreme high and low temperatures tend to occur over the total record, and during the year?

- 2) Let's focus on the 3 months of the year with the greatest number of extreme high daily temperatures, which from Figure 3 should be July, August, and September

(JAS). Plot a histogram of JAS temperature (normalized to a pdf) over the first 30 years of the record, and overplot the same histogram computed over the last 30 years (**Figure 4**).

Q2: How do the two histograms differ?

- 3) Compute the 10th, 50th, and 90th percentiles for each JAS day using all years. Plot the percentiles vs day of the summer (**Figure 5**). Overplot temperature for the summer of 2022.
- 4) Compute the number of days each summer when temperature exceeds the 95th percentile. Plot this as an indicator of the number of summer heatwave days (**Figure 6**).

Q3: How has the number of heatwave days changed over the record?

- 5) Compute the mean JAS temperature for each year and plot this as a time series. Use *plotyy* (**Figure 7**) and scatter (**Figure 8**) to compare number of heatwave days and the mean JAS temperature.

Q4: How do changes in the number of summer heat wave days compare to changes in the mean JAS temperature?

Assignment 1 - Scripps Pier Temperature

Figure 1. Plot of SIO Pier temperature, with extreme high values (> 99th percentile of entire record) indicated in red

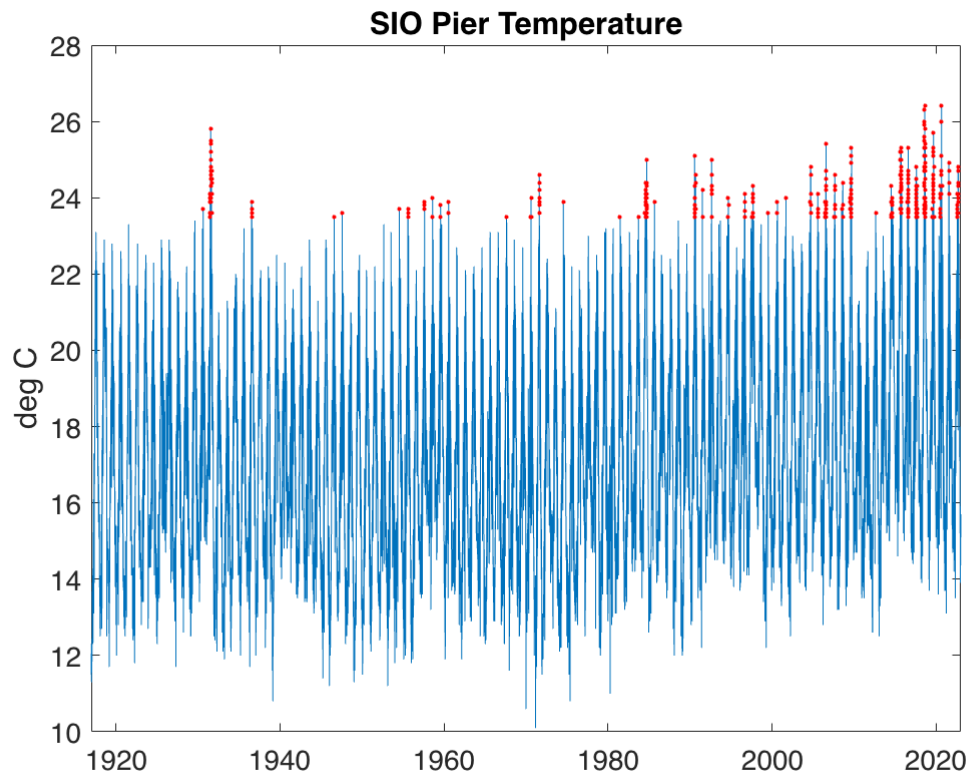


Figure 2. Histogram of year of high values

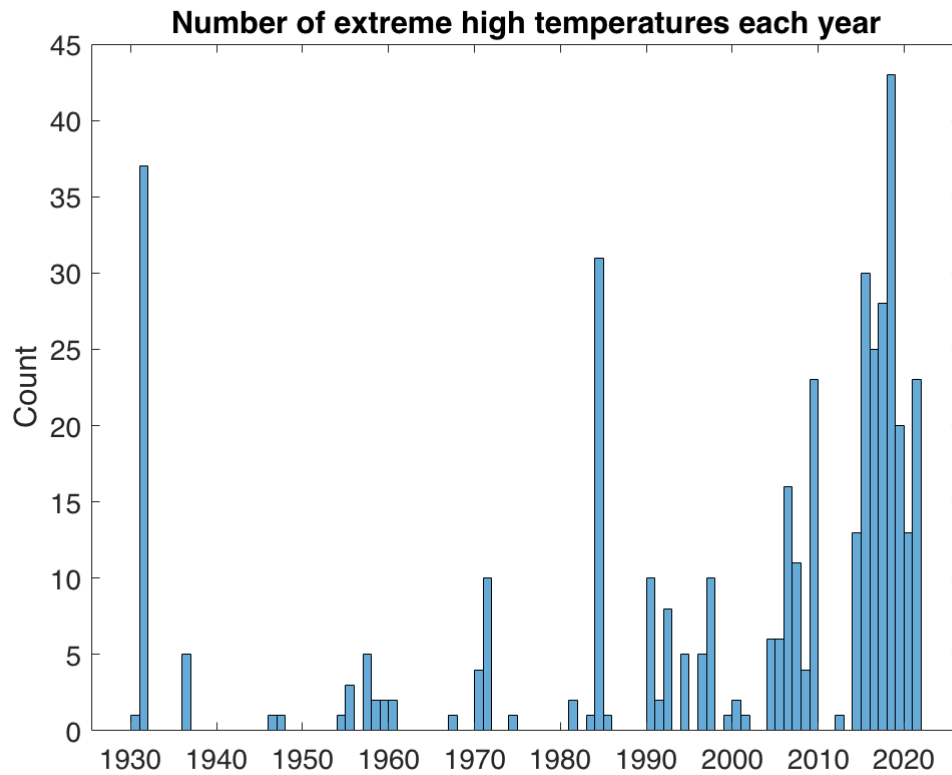


Figure 3. Histogram of month of high values

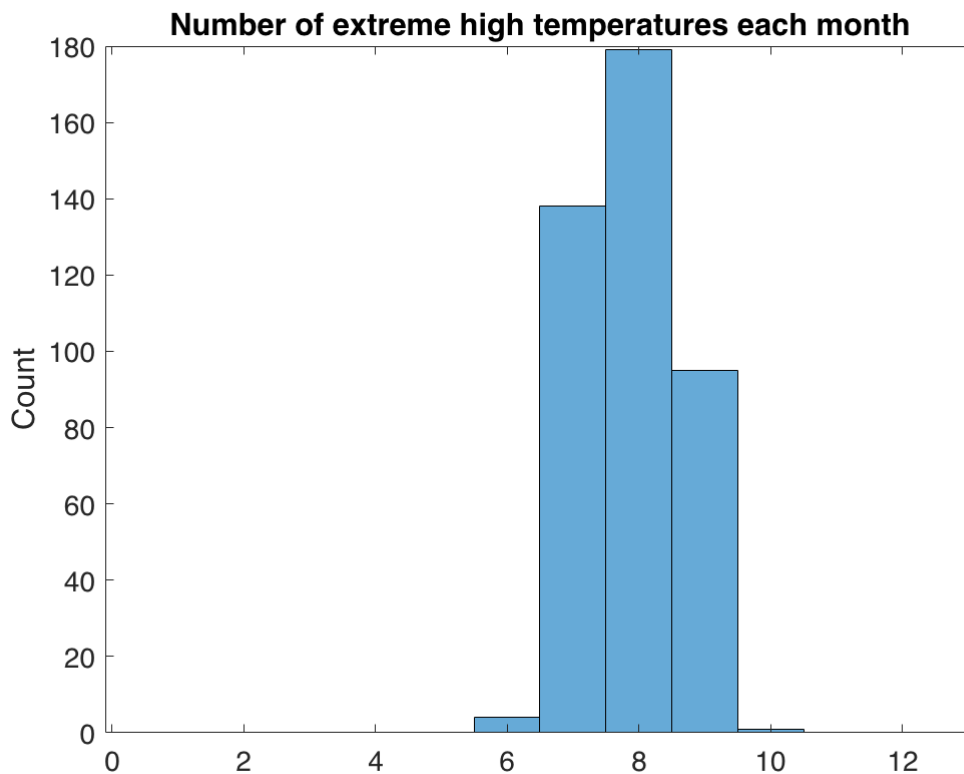


Figure 4. Plot of SIO Pier temperature, with extreme low values (< 1 percentile of entire record) indicated in red

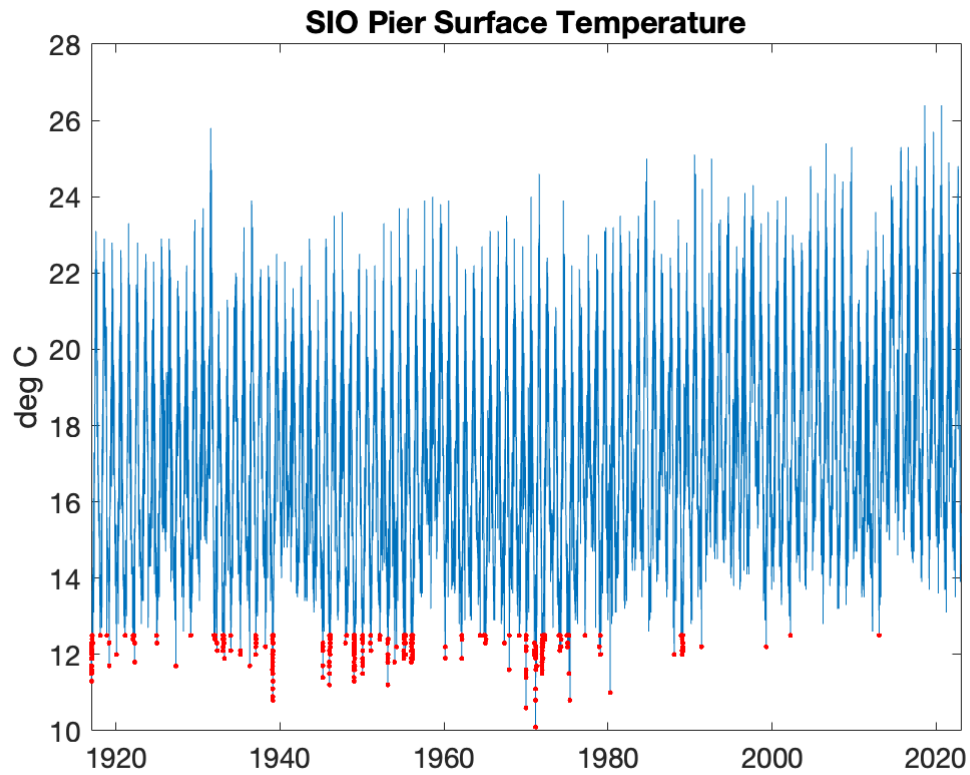


Figure 5. Histogram of year of low values

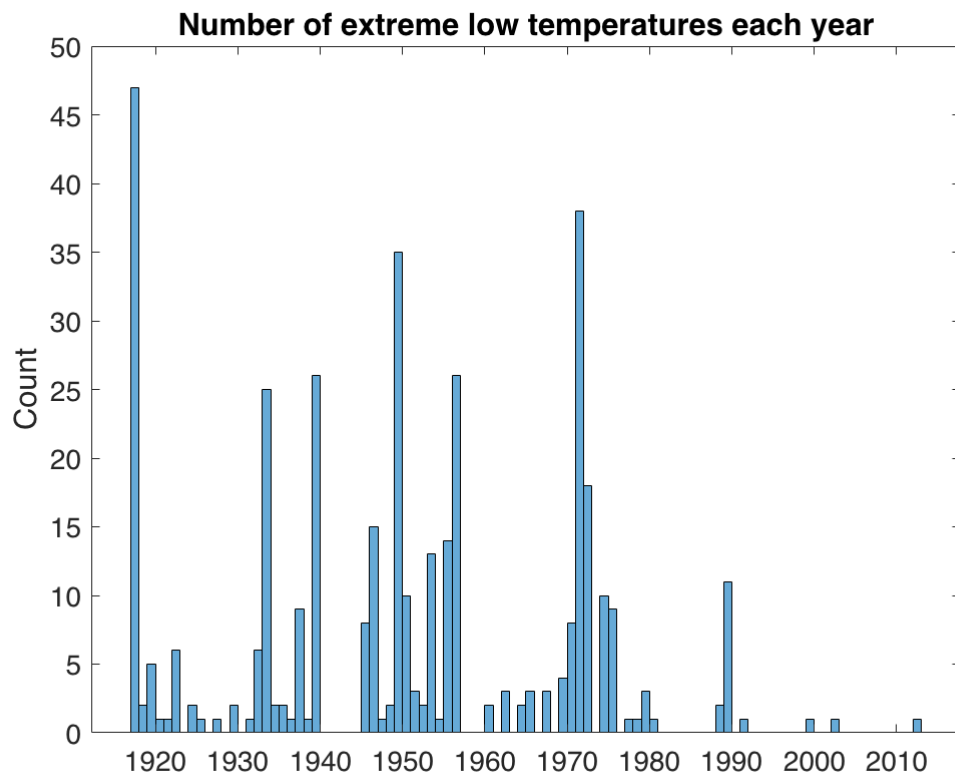
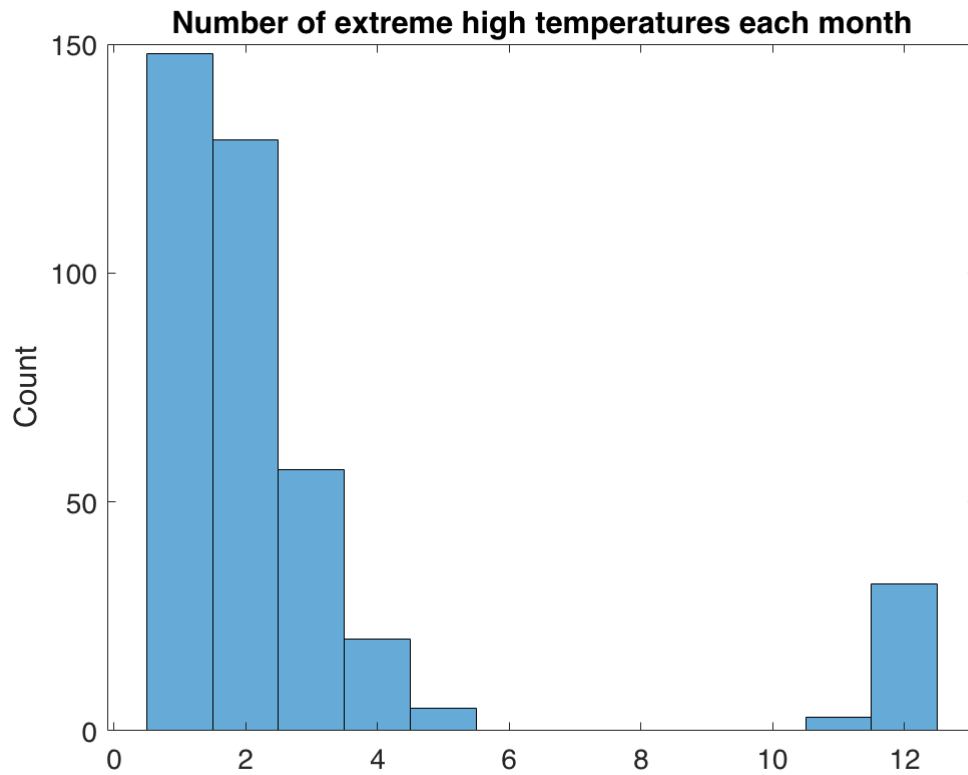
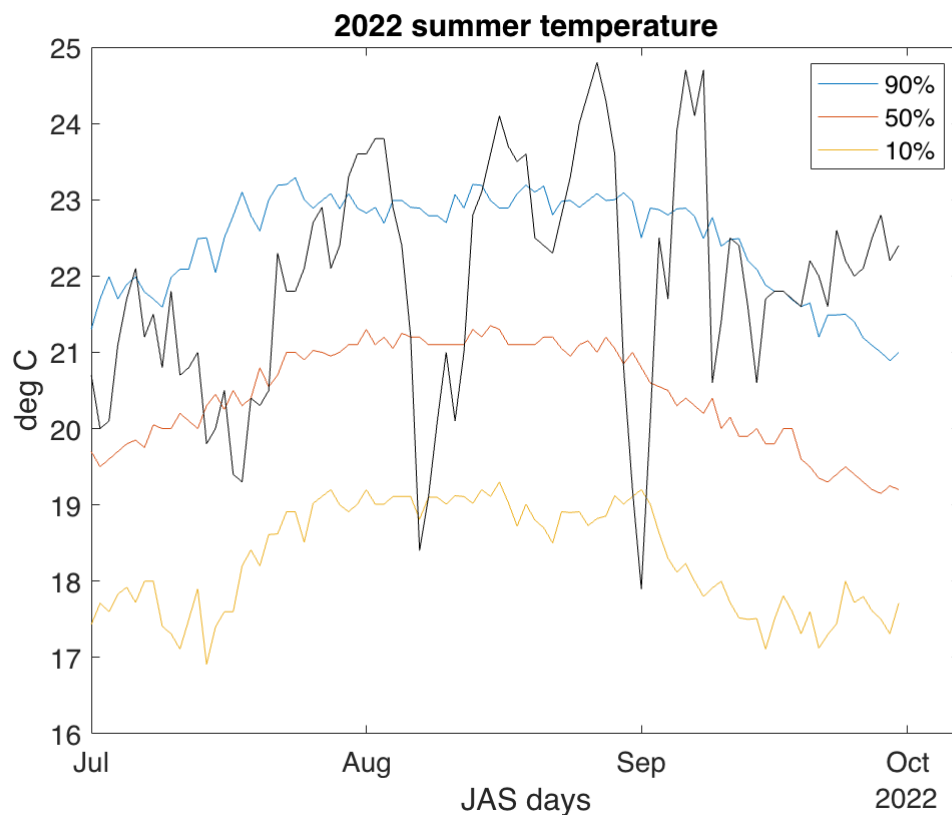


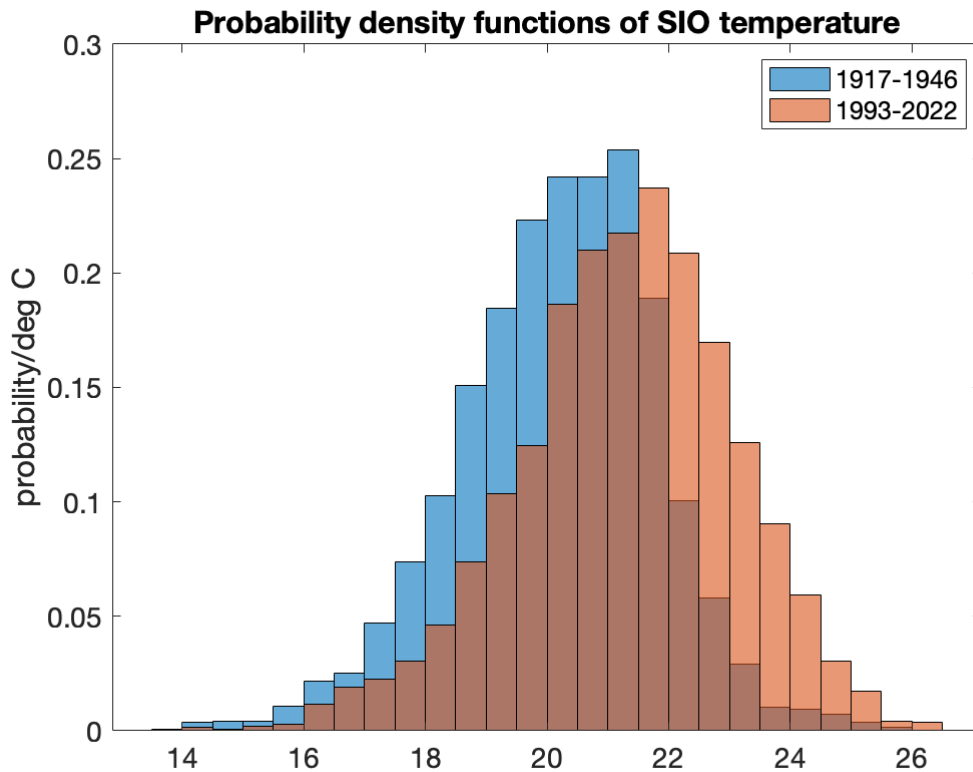
Figure 6. Histogram of month of low values



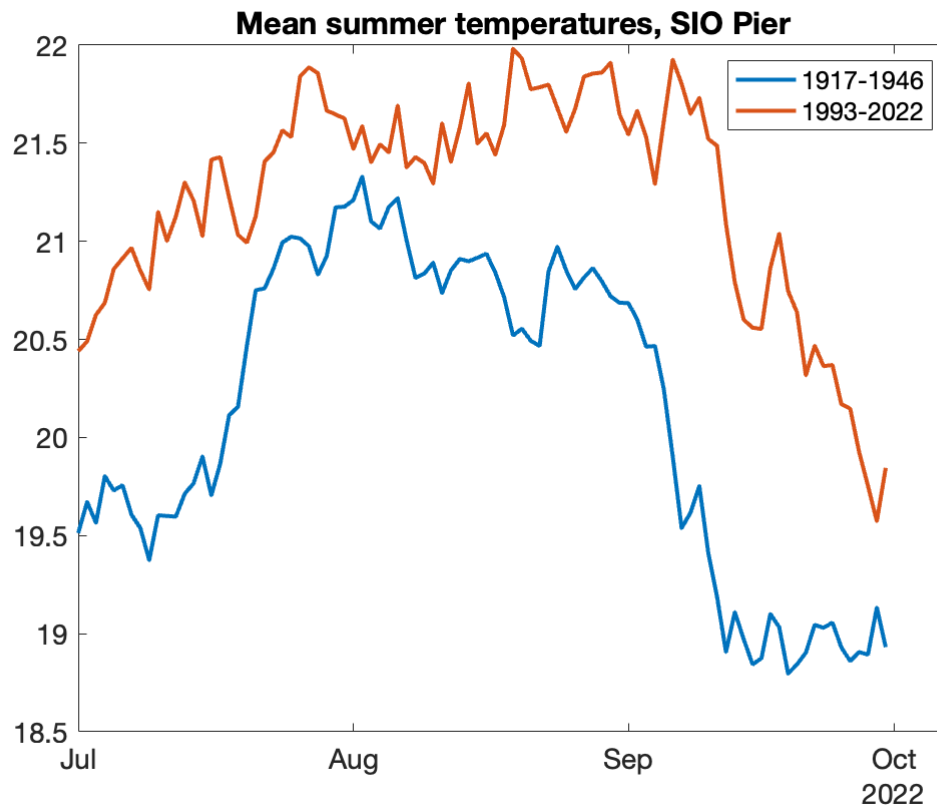
7. JAS summer temperatures during 2022, with 10th, 50th, and 90th percentiles



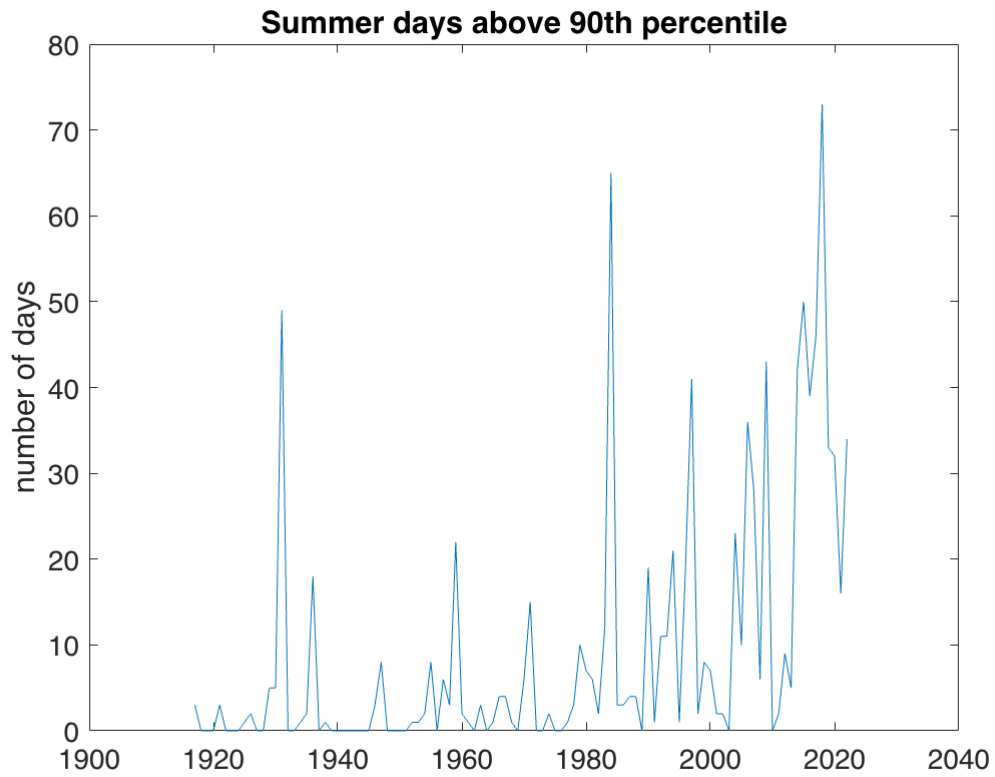
8) Histograms of temperature during first 30 and last 30 years



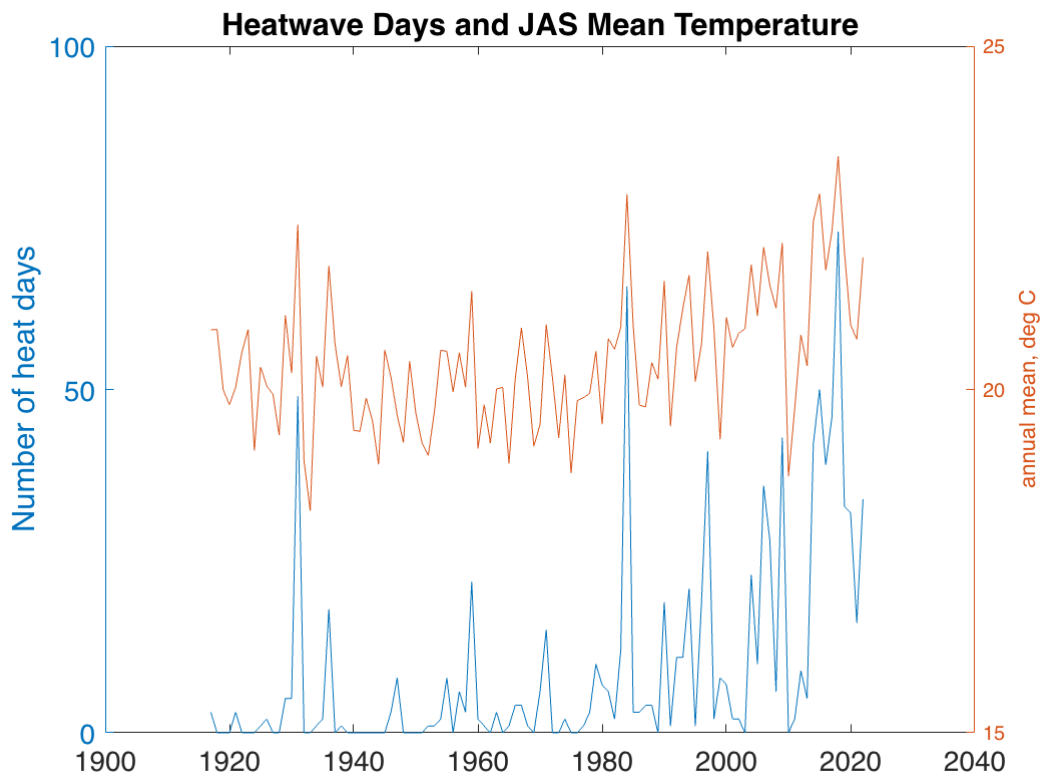
9) Time series of average JAS summer temperatures during first 30 and last 30 years



10) Number of JAS days above the 90th percentile



11) Time series plot of mean JAS temperature and heat wave days



12) Scatter plot of mean JAS temperature vs heat wave days

Mean JAS temperature vs Heatwave days

